Assignment 2 Report

ChordCast: Scalable P2P Task Execution & Gossip Engine

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1. Introduction

This project implements a **Peer-to-Peer (P2P) distributed system simulation** using OMNeT++. The system models a network of nodes arranged in a ring topology and enhanced using **CHORD-based routing** for efficient communication.

Additionally, the system supports:

- Distributed task execution
- Efficient message routing
- Gossip-based dissemination with duplicate suppression

The objective is to simulate **scalable, decentralized computation and communication** in a structured overlay network.

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2. Objectives

- Design a **ring-based P2P network**
- Optimize routing using **CHORD protocol (O(log N))**
- Implement **distributed task execution**
- Develop a **gossip protocol** for message propagation
- Ensure correctness, efficiency, and termination

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3. System Architecture

The system consists of:

Components:

- **ClientNode**
  - Handles routing, execution, and gossip
- **NetworkController**
  - Initializes topology and network configuration

Key Layers:

- **Topology Layer** → Ring structure
- **Routing Layer** → CHORD finger tables
- **Execution Layer** → Task distribution & computation
- **Communication Layer** → Gossip protocol

Screenshot 1: Network Initialization

(Insert screenshot showing ring topology logs here)

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4. Network Topology

- Nodes are connected in a **bidirectional ring**
- Each node maintains:
  - Successor
  - Predecessor

This ensures:

- Connectivity
- Fault tolerance (basic level)

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5. CHORD-Based Routing

To optimize routing, each node maintains a **finger table**:

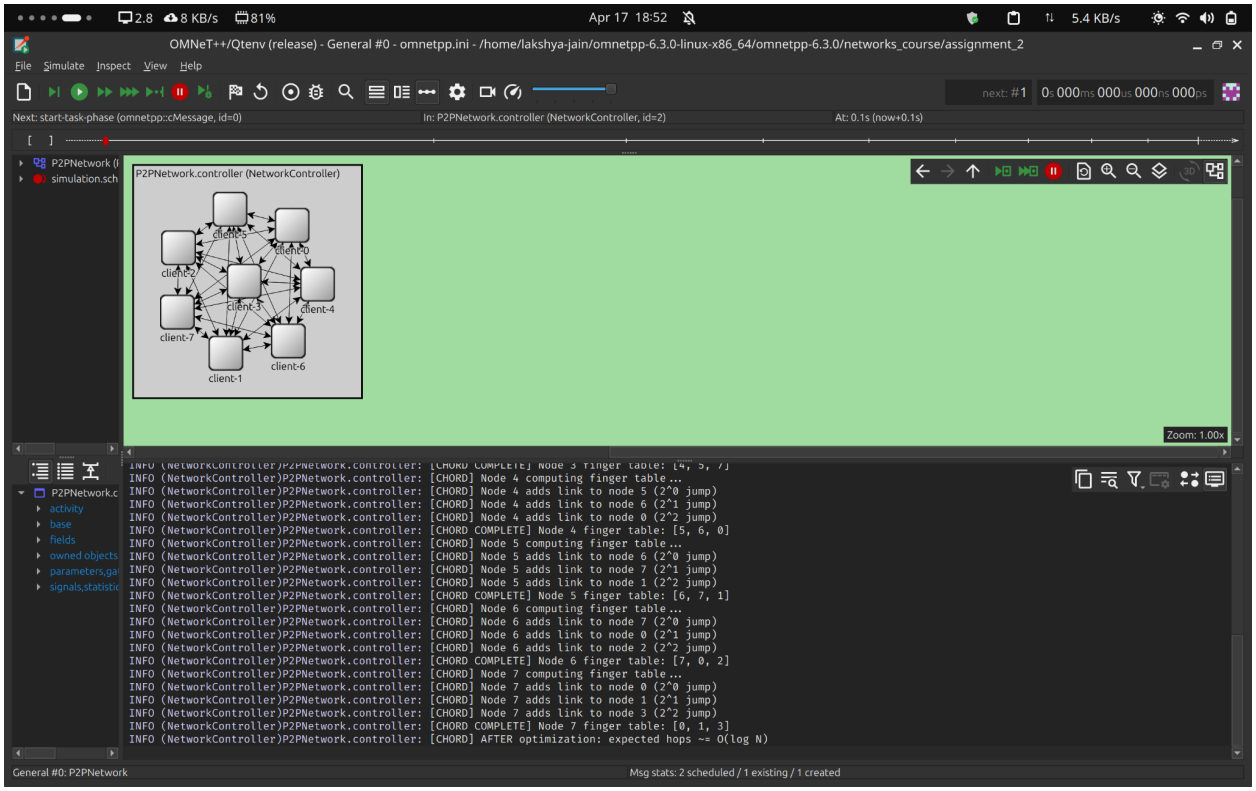
finger[i] = (node + 2^i) % N

Benefits:

- Reduces lookup complexity:
  - From **O(N) → O(log N)**
- Enables long-distance jumps in the network

Routing Strategy:

- Select **closest preceding node** toward target
- Avoid linear traversal



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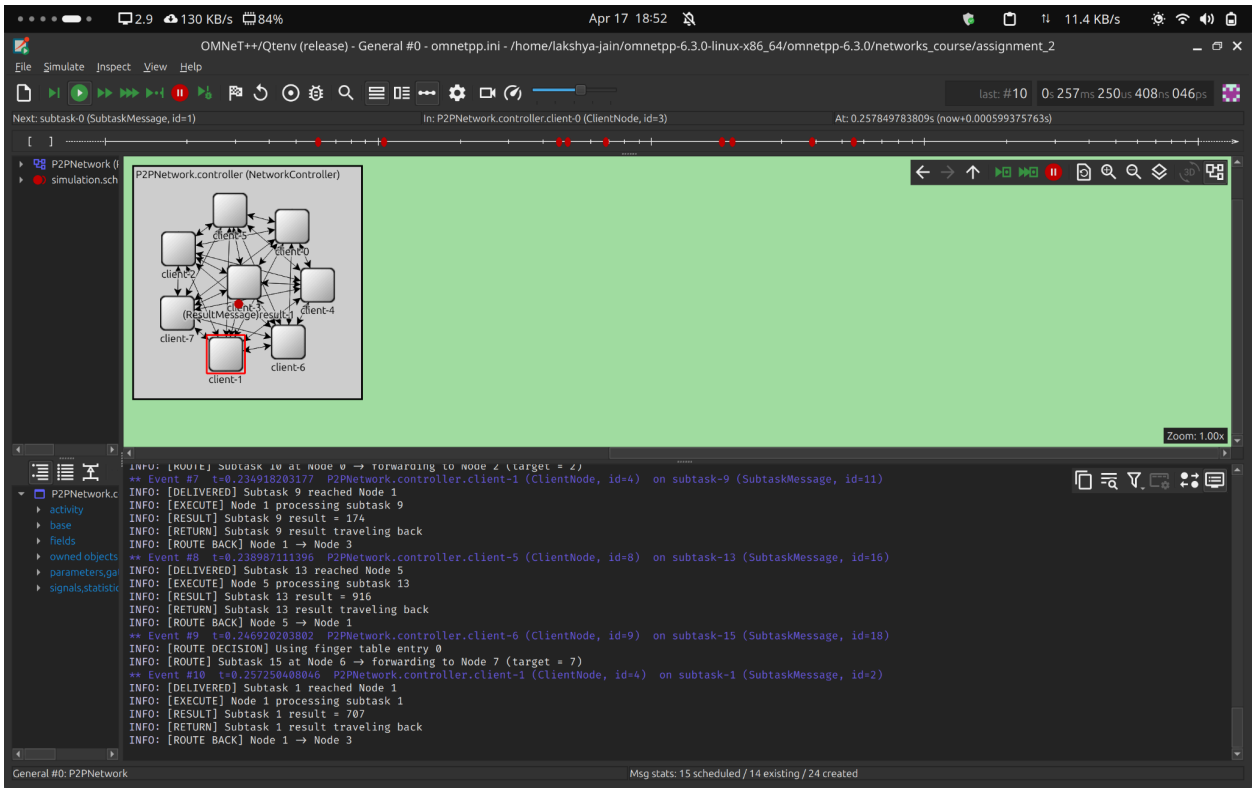
6. Distributed Task Execution

Process:

1. A node is selected as **initiator**
2. Task is divided into **subtasks**

Each subtask is assigned:
client = subtask_id % N

3.
4. Subtasks are routed using CHORD
5. Target node executes and returns result
6. Initiator aggregates results



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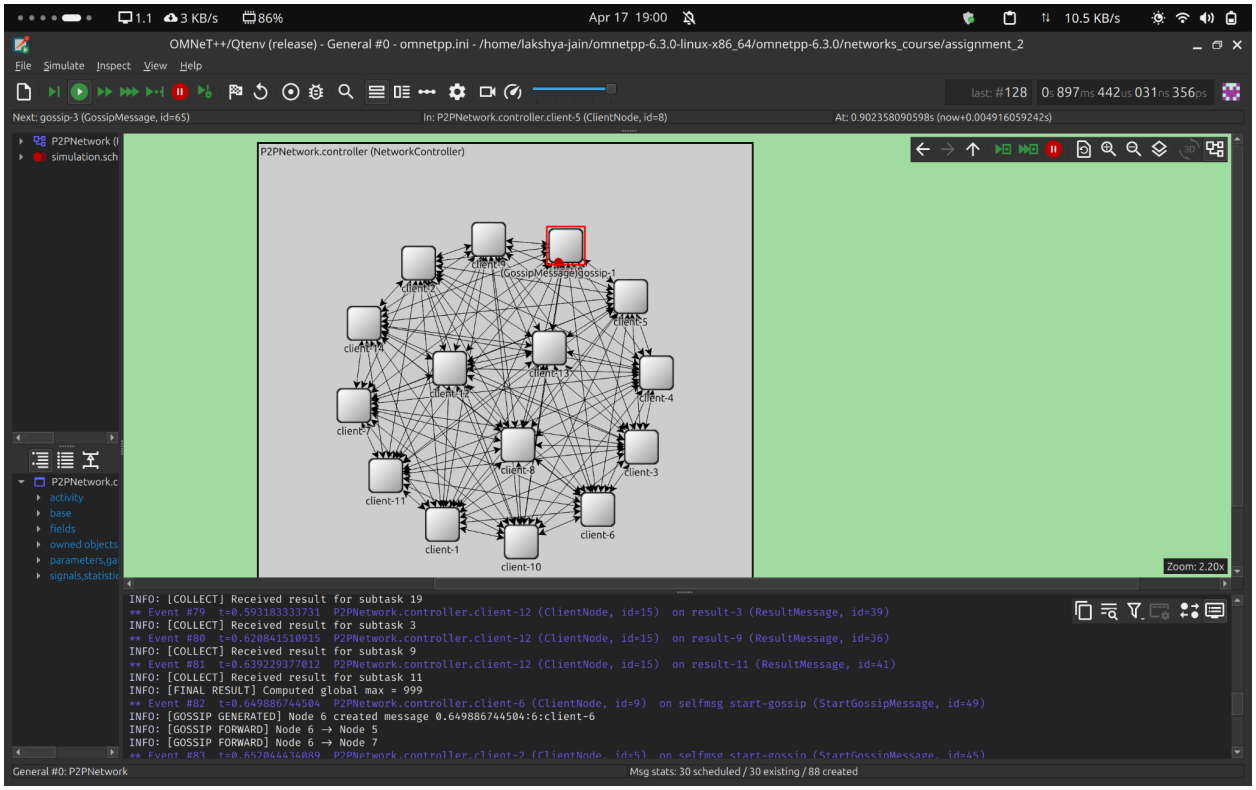
7. Result Aggregation

- Each node computes a **local result**
- Results are routed back to initiator
- Final result is computed (e.g., max value)



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8. Gossip Protocol

After computation, nodes initiate **gossip dissemination**.

Message Format:

<timestamp>:<node_id>:<client_id>

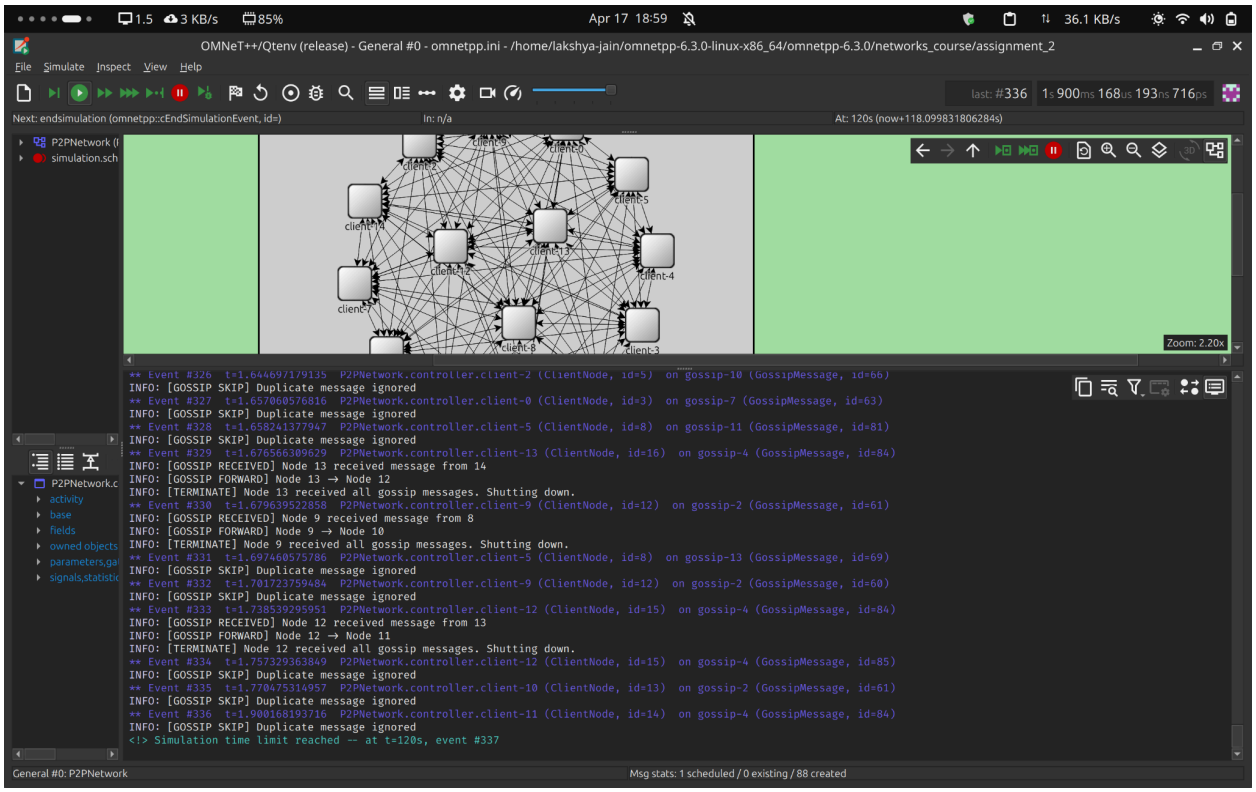
Rules:

- Forward only to **neighbors**
- Do not send back to sender
- Maintain **Message List (ML)** to avoid duplicates
- Ignore already seen messages

Workflow:

1. Node generates gossip message

2. Checks ML:
    - If new → store and forward
    - If duplicate → drop
3. Message propagates across network
4. Node terminates after receiving all messages



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9. Complexity Analysis

Operation	Complexity
Ring Routing	O(N)
CHORD Routing	O(log N)
Task Distribution	O(N)
Gossip Propagation	O(N)

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10. Key Achievements

- Implemented **efficient CHORD routing**
- Achieved **distributed computation pipeline**
- Designed **robust gossip protocol**
- Eliminated:
    - Broadcast storms
    - Duplicate flooding
- Ensured correct **termination conditions**

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11. Challenges Faced

- Correct implementation of **CHORD routing logic**
- Avoiding **duplicate message propagation**
- Ensuring **no infinite gossip loops**
- Handling **event scheduling constraints in OMNeT++**

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12. Improvements & Future Work

- Dynamic node join/leave (full CHORD implementation)
- Fault tolerance and recovery
- Load balancing across nodes
- Visualization of routing paths
- Performance benchmarking for large N

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13. Conclusion

This project successfully demonstrates a **scalable distributed system** using:

- Structured overlay networks
- Efficient routing mechanisms
- Decentralized communication protocols

The integration of **CHORD + Gossip** provides a strong foundation for real-world distributed systems such as:

- Distributed hash tables

- Blockchain networks
- Peer-to-peer communication systems